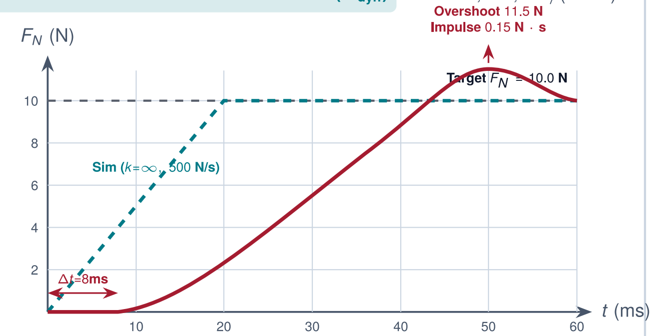
THE MULTIDIMENSIONAL SIM-TO-REAL REALITY GAP VECTOR $\boldsymbol{\Delta} = [\Delta_{\mathrm{dyn}}, \Delta_{\mathrm{lat}}, \Delta_{\mathrm{sens}}, \Delta_{\mathrm{vis}}]^T$

Task-relevant mismatches across heterogeneous physical channels: each carries distinct units, failure modes, runtime detectors, and unclosable residuals.

CHANNEL 1: DYNAMICS & CONTACT (Δ_{dyn})

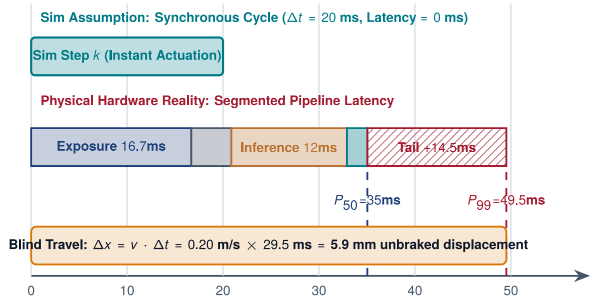
Units: N, N/s, rad/(N · m)



Physical Failure Signature: Delayed force rise interpreted as non-contact → commands extra motor torque → 3.0 mm over-travel into current trip (I_{max}).
Runtime Detector: Torque/force derivative clamp ($dF/dt \leq 2.5 \times 10^4$ N/s) in MCU reflex loop.
Unclosable Residual: Gearbox compliance (1.5×10^{-3} rad/N · m) & contact damping.

CHANNEL 2: ACTUATOR & LATENCY (Δ_{lat})

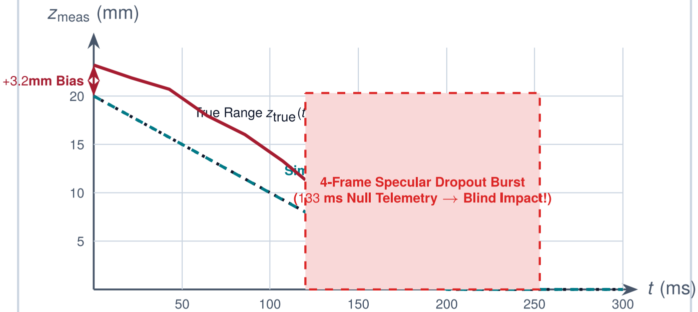
Units: ms, mm



Physical Failure Signature: Commands arrive 29.5 ms late → 5.9 mm displacement depletes phase margin → high-frequency limit-cycle oscillation and impact shocks.
Runtime Detector: Hardware timestamp age watchdog and RTOS jitter monitor.
Unclosable Residual: Inference contention tail (P_{99} = 49.5 ms, jitter ±6.2 ms).

CHANNEL 3: SENSING & TRANSDUCTION (Δ_{sens})

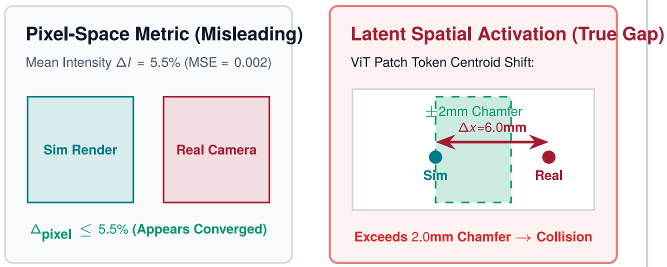
Units: mm, N, lux



Physical Failure Signature: +3.2 mm bias triggers premature regulation 3.2 mm above surface; 133 ms optical dropout blinds policy during descent → unbraked crash.
Runtime Detector: Consecutive dropout frame counter and thermal drift estimator.
Unclosable Residual: Thermal drift (0.40 N/10°C), noise floor (σ = 1.8 mm).

CHANNEL 4: PERCEPTUAL / VISUAL (Δ_{vis})

Units: Cosine Dist, mm



Physical Failure Signature: 5.5% pixel loss masks 6.0 mm spatial token offset → end-effector strikes fixture shoulder instead of bore.
Runtime Detector: Feature-space Mahalanobis distance monitor on encoder tokens.
Unclosable Residual: Specular BRDF reflection and micro-texture variations.

Systems Architectural Synthesis:
The sim-to-real gap is inherently a 4D coupled vector $\boldsymbol{\Delta} = [\Delta_{\mathrm{dyn}}, \Delta_{\mathrm{lat}}, \Delta_{\mathrm{sens}}, \Delta_{\mathrm{vis}}]^T$. Sizing the gap requires paired-trace telemetry across each physical axis. Domain randomization broadens coverage over modeled parameters (Ξ) but cannot bridge structural omissions or unmodeled timing tails outside the simulator graph.